Consumer Demand for Food

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From Minimum Cost to Actual Choices

- In Project 2 we asked: what *should* people eat?
  - Find the cheapest diet meeting nutritional requirements.
- In Project 3 we ask: what *do* people eat?
  - And how do their choices respond to prices and income?

Key Questions

- How do actual diets compare with minimum cost diets?
- How do changes in food prices affect what people eat?
- How do changes in income affect what people eat?
- Can prices and incomes ensure people choose nutritious diets?

Consumers Make Choices

Most of us do not solve a linear program at the grocery store. Instead:

- We have a *budget* — how much we can spend.
- We face *prices* — what each food costs.
- We have *preferences* — what we like to eat.

The consumer's problem: choose the *best* bundle of goods you can afford.

An Everyday Example

Suppose you have \$10 for lunch. You can buy:

- Burritos at \$5 each
- Salads at \$4 each

What do you choose? It depends on how much you *like* each option relative to its price.

Making This Precise

Economists write the consumer's problem as:

$$\max_c U(c) \quad \text{subject to} \quad p'c \leq x$$

- c the *consumption bundle* (quantities of each food)
- p the *price vector* (price per unit of each food)
- x the *budget* (total money available)
- $U(c)$ *utility* — a number measuring how satisfied you are

What Is Utility?

- A way to *rank* bundles: if $U(A) > U(B)$, you prefer bundle A.
- Not a measure of "happiness units" — only the ranking matters.
- We assume utility is:
  - *Increasing* — more food (weakly) preferred to less.
  - *Concave* — variety is good; diminishing returns to any single food.

Indifference Curves

An *indifference curve* connects all bundles that give the same utility.

- Every point on the curve is equally good.
- Curves further from the origin are better.
- Curves never cross (why?).

The Budget Line

The budget line shows all bundles that exactly exhaust your budget:

$$p_1 c_1 + p_2 c_2 = x$$

Slope $-p_1/p_2$ — the rate at which the market lets you trade one good for another.

Intercepts x/p_1 on the c_1 axis, x/p_2 on the c_2 axis.

Finding the Best Choice

The consumer picks the bundle on the *highest* indifference curve that still touches the budget line.
At the optimum, the indifference curve is *tangent* to the budget line:

$$\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$$

Left side your *marginal rate of substitution* (MRS) — how much c_2 you'd give up for one more unit of c_1.

Right side the *price ratio* — how much c_2 the market requires you to give up.

Cobb-Douglas Utility

A simple and widely-used functional form:

$$U(c_1, c_2) = \alpha_1 \log c_1 + \alpha_2 \log c_2$$

- $\alpha_1, \alpha_2 > 0$ measure relative preference for each good.
- The log gives diminishing marginal utility — eating 10 burritos is not 10 times as good as eating 1.

Demands Under Cobb-Douglas

Solving the consumer's problem yields a clean result:

$$c_i^* = \frac{\alpha_i}{\alpha_1 + \alpha_2} \cdot \frac{x}{p_i}$$

Each good gets a *fixed share* of the budget.
Example: if $\alpha_1 = 0.3$ and $\alpha_2 = 0.7$, then 30% of income goes to good 1 and 70% to good 2, no matter what prices are.

Is Constant Budget Shares Realistic?

Think about your own spending as income rises:

- Do you spend the same *fraction* on rice? On restaurant meals?
- Probably not — this is a limitation of Cobb-Douglas.
- We'll need richer models for the project.

Demand Curves

Fix income and all other prices. Vary p_1:

- As p_1 rises, you buy less of good 1 (usually).
- The plot of c_1^* against p_1 is the *demand curve*.
- Convention (blame Alfred & Mary Paley Marshall): price on the vertical axis!

What Shifts a Demand Curve?

The demand curve for good 1 shifts when:

- *Income* changes — you can afford more or less.
- *Other prices* change — substitutes become cheaper or more expensive.
- *Preferences* change — new information about health, taste, etc.

Engel Curves

Fix all prices. Vary the budget x:

- The plot of c_i^* against x is the *Engel curve* (named for Ernst Engel, one of your P3 team names!).

Normal goods have demand that rises with income. *Inferior goods* have demand that falls with income (e.g., instant noodles).

Engel's Law

Ernst Engel (1857) observed: as income rises, the *share* of spending on food falls.

This is one of the most robust empirical findings in economics:

- Poor households spend 50–80% of income on food.
- Rich households may spend 10–15%.
- This has big implications for how price changes affect welfare.

Why We Need More Flexible Models

Cobb-Douglas forces constant budget shares. Real data shows:

- Budget shares change with income (Engel's Law).
- Some goods are substitutes; others are complements.
- Responses to price changes differ across foods.

Constant Frisch Elasticity (CFE) Utility

A more flexible form:

$$U(c) = \sum_{i=1}^{n} \alpha_i \frac{\beta_i}{\beta_i - 1} \left(c_i^{1-1/\beta_i} - 1 \right)$$

- α_i still governs relative importance of each good.
- β_i controls the *curvature* — how quickly marginal utility diminishes.
- When all $\beta_i = 1$, this reduces to Cobb-Douglas.

What Changes?

With different β_i values:

- Budget shares now depend on prices and income.
- Engel curves can be nonlinear.
- Demand curves can have different shapes for different goods.

This is closer to the kind of model you'll estimate in your project.

What Your Team Will Do

1. Choose a real-world population with expenditure data.
2. Estimate a demand system — how food choices respond to prices and income.
3. Use the estimates for counterfactual experiments:
   - What if incomes doubled?
   - What if food prices rose 20%?
   - Would people still get adequate nutrition?

From Theory to Data

Today's theory gives you the framework. Your project will:

- Replace the simple 2-good model with many food categories.
- Use real household survey data instead of hypothetical numbers.
- Estimate the α and β parameters from observed choices.

The notebook `food_demand.org` walks through a computed example — we'll look at that next time.